

LANESKIFF · REQUIREMENTS

LaneSkiff requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/laneskiff		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Rated load and test conditions stated; TRL 3 status from LSK-CAL-001; targets unchanged
0.1	2026-09-30	Amish Chadha	Initial scaffold

LaneSkiff requirements

Requirements for the first prototype, built on the plywood and epoxy path. Targets are those set at TRL 1 and are unchanged. Status is the paper result at TRL 3 from LSK-CAL-001; every requirement is verified by test at TRL 4. Where a requirement is not met or is at risk on paper, the state and the options are set out for Amish in `docs/REVIEW.md` (TRL 3 section) and listed as open decisions in LSK-DEC-001.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 4)	TRL 3 status
R1	Carry mass	Complete hull 50 kg or less; each half 25 kg or less (plywood and epoxy path, target)	Weigh the prototype halves	Not met on paper: hull 100 kg, halves 56 and 45 kg (decision for Amish)
R2	Payload	At least 6 times hull mass, about 300 kg or four adults plus crew (target)	Static load test with water ballast in calm water	Not met on paper: 450 kg rated load is 4.5 times hull mass (decision for Amish)
R3	Draft at rated load	15 cm or less	Measured at the transom and bow during the load test	Not met on paper: 187 mm at the forward end of the waterline, 162 mm at the transom, six persons (decision for Amish)
R4	Width for lanes	Beam 1.2 m or less overall	Tape measure; trial passage through a 1.3 m gate	Met: 1,137 mm over the rub strakes
R5	Level flotation when swamped	Floats approximately level with rated persons aboard when fully swamped, using the ABYC H-8 test method (not certified)	Swamp test in a pool following ABYC H-8 procedure	Met on paper with a small margin: 2.1 deg trim, 36 mm freeboard at the transom (at risk; decision for Amish)
R6	Boarding from waist-deep water	An 80 kg adult boards over the stern step unaided; heel stays under 10 degrees (target)	Boarding trials with several volunteers of different builds, inclinometer on the hull	Met on paper: 4.3 deg heel over the stern step (15.6 deg over the side for comparison)
R7	Assembly	Two halves joined in 10 min or less by two people with one spanner	Timed assembly trial	Met on paper (estimate): about 6 min, one 17 mm spanner
R8	Local build	At least two build paths (plywood and epoxy, aluminium or plastic sheet) documented with mass and payload	Build by a partner boatyard from the published drawings	Met on paper: three paths in the material substitution table (LSK-CAL-001, section 8)
R9	Prototype cost	USD 1,500 or less including step, poles and paddles	Bill of materials and receipts	USD 1,922: over the value-engineering target by USD 422
R10	Optional fin drive	Removable in 5 min or less; the boat meets R2 to R6 without it	Trials with and without the drive fitted	Not shown on paper: no drive in the first prototype design (decision for Amish)

■ Definitions used for the TRL 3 status

- **Hull mass:** the complete boat with its stern step, handles, strakes, skids and buoyancy, as carried; the push pole, paddles and bow line are loose gear (6.2 kg) and not counted.
- **Rated load:** six persons at 75 kg (a crew of two and four adults, 450 kg), placed as in LSK-CAL-001, section 3. The final rating of the first prototype is part of the R3 decision.
- **Swamped:** both halves full to the outside water level, persons aboard counted at two thirds of their weight (a conservative design assumption, not the ABYC H-8 rule itself).
- **Draft:** from the water surface to the outside of the bottom, at the transom and at the forward end of the waterline.

■ Assumptions

- Fresh floodwater, 1,000 kg/m³; calm, slow water.
- Marine plywood of about 600 kg/m³ and durable hardwood of about 700 kg/m³, as found in India and Bangladesh; lighter okoume plywood is an option (LSK-CAL-001, section 9).
- Closed-cell polyethylene foam of about 30 kg/m³ with under 1 % water uptake by volume is available near the first partner.
- Two people can carry about 40 to 50 kg between them over a short distance; four can carry the assembled boat.
- A stern-centre step heels the boat less than a side step; LSK-CAL-001 confirms this on paper, the boarding trials confirm it in water.

Safety: LaneSkiff is an open engineering reference for a rescue boat, not certified equipment. Every trial behind R2, R3, R5 and R6 is run in calm, shallow water with a safety boat or bank team, every person in a life jacket, and the volunteers briefed; no trial uses moving water. The step hinge and stop straps are proof-loaded to twice their working load before any boarding trial.